

MATH 210 — Problem Set 4

1. Evaluate the definite integral:

$$\int x^2 dx \text{ from } 0 \text{ to } 3$$

2. Simplify, assuming $n \geq 1$:

$$\sum (2k \pm 1) \text{ for } k = 1 \text{ to } n$$

3. Solve for x where $x \neq 0$:

$$\sqrt{x^2 + 16} \leq 5$$

4. Compute the partial derivative:

$$\partial f / \partial x \text{ where } f = 3x^2 y - y^3$$